# Digital Twins for Predictive Maintenance in Automotive Aftersales

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***Abstract*—This paper investigates the effect of digital-twin-based predictive maintenance on the automotive aftersales domain, focusing on the German passenger-car market. A model-based gap analysis is conducted to compare the interval-based current state against a digital-twin-enabled target state along the full customer journey. Based on this comparison, the business potential is quantified using a transparent assumption framework anchored in published German market data. The results show that proactive, data-driven service recommendations raise the repair capture rate from 40 to 70 percent, increasing wear-part gross profit by approximately 53 EUR per vehicle per year and 263,000 EUR annually across a 5,000-vehicle fleet. Customer touchpoints increase by up to 1.1 per year, from 1.6 to roughly 2.7, each representing an additional advisory and upselling opportunity. The analysis further identifies data access, rather than technology maturity, as the dominant adoption barrier. Together, these results establish a transparent, contestable model of the potential that dealerships and original equipment manufacturers can use to evaluate and plan the integration of digital-twin-based predictive maintenance into their aftersales operations.**

***Index Terms*—Digital Twin, Predictive Maintenance, Connected Vehicle, Machine Learning, Automotive Aftersales**

## I. Introduction

Today's vehicles are connected by default and continuously transmit operational data [1], [2], so the information required to assess their technical condition is largely available. Nevertheless, automotive aftersales remains reactive: even well-connected vehicles are serviced on fixed time or mileage intervals rather than on the actual condition of their components [3]. Wear parts—brake pads and discs, tires, ignition components, and wipers—are typically replaced only after wear or failure, leading to unplanned repairs frequently carried out by independent third-party workshops [4]. For the dealership, which owns the customer relationship, this means losing repair revenue and valuable interactions, reducing opportunities for customer advice, retention, and upselling [5], [6].

Digital-twin-based predictive maintenance (PdM) can turn this data into foresight. A digital twin (DT) maintains a synchronized virtual representation of the vehicle [7], against which machine-learning (ML) models forecast the remaining useful life (RUL) of components and recommend replacement before failure—strengthening customer retention at the dealership. While the building blocks of DT-based PdM are well studied in industrial settings [8], [9], their integration into automotive aftersales remains underexplored.

This paper therefore addresses the following research question: *what effect does the proactive use of digital twins and predictive maintenance have on the automotive aftersales domain?* Focusing on the German market and working with explicitly stated assumptions, the study (i) clarifies the interplay of DT and PdM in aftersales, (ii) compares the predictive target state with the interval-based current state along the full customer journey to identify where reactive servicing causes value leakage, and (iii) derives actionable recommendations for dealerships and original equipment manufacturers (OEMs).

## II. Related Work and Theoretical Background

### A. Digital Twins

The concept of the digital twin was formalized by Grieves [7] in a white paper on manufacturing excellence, where it described a virtual replica of a physical product maintained in synchrony with its real-world counterpart throughout the entire asset lifecycle. Since then, the concept has expanded into aerospace, energy, and automotive domains [8], [10].

In the automotive domain, this general architecture maps directly onto the connected vehicle. The physical entity is the vehicle itself, equipped with on-board sensors and an On-Board Diagnostic (OBD) interface. Its virtual representation is a cloud-hosted model mirroring component states, degradation histories, and operating conditions. The synchronizing link is the over-the-air (OTA) telematics connection, which transmits telemetry data to the cloud and returns configuration updates to the vehicle [1], [2], [11]. Together, this setup transforms the vehicle from a periodically inspected asset into a continuously monitored system—a shift first documented for automotive applications by Zhang et al. [11].

### B. Vehicle Maintenance Strategies

Vehicle maintenance strategies form a maturity hierarchy: reactive maintenance repairs only after failure [9]; preventive maintenance schedules service at fixed mileage or time intervals, leading to either premature replacement or undetected wear [9], [12]; condition-based maintenance (CBM) couples the service decision to sensor thresholds [12]; and PdM extends CBM by forecasting the future evolution of degradation, estimating the remaining time or distance until failure [9], [13].

In this architecture, PdM is not a stand-alone technique but a service layer built on a DT of the vehicle. As described by Lopes et al. [1] and El Hadraoui et al. [12], the DT provides the synchronized virtual representation against which sensor streams are interpreted, anomalies detected, and RUL estimates computed. In current automotive DT architectures, predictive analytics is therefore typically positioned as a dedicated microservice within the DT's analytics core [1], [14].

### C. Machine Learning in Predictive Maintenance

Three main problem classes structure ML-based PdM. Classification assigns sensor patterns to discrete fault categories using Support Vector Machines, Random Forests, or Convolutional Neural Networks (CNNs) on time-frequency representations of vibration signals [9], [13]. Regression targets RUL estimation, dominated by Long Short-Term Memory (LSTM) networks for their ability to model long-range temporal dependencies in degradation signals [9], [14]. Anomaly detection operates without explicit labels, learning nominal behavior and flagging deviations as failure precursors [9], [14]. AbouGrad et al. report classification accuracies of approximately 89% for vibration-based fault detection and 92% for LSTM-based RUL estimation in automotive contexts, though they note that these figures depend strongly on dataset quality and operating conditions [9]. In the aftersales context, RUL regression is the primary mechanism behind proactive service recommendations, whereas anomaly detection serves as an early-warning layer for components without established wear curves.

### D. Automotive Aftersales

Aftersales encompasses post-purchase activities that enable customers to access supplier services and resolve product issues through continuous maintenance [15]. In the automotive context, it represents a dedicated business function comprising maintenance, repair, and warranty offerings to maximize product usage over the entire asset lifecycle while meeting economic and customer-related objectives [6], [16]. Building post-sales relationships strengthens customer loyalty, yields actionable insights, and creates upselling and cross-selling opportunities, making the aftersales market increasingly attractive as demand for higher-quality service expands [6].

Historically, once the warranty expires, customer engagement with the dealership tends to decline [17]. Within the automotive industry, aftersales services represent a significant source of revenue, with profit margins substantially higher than those generated through vehicle sales [18]. From an economic perspective, the aftersales market has been identified as larger than the primary product sales market across various industries [10], [19], [20]. Retaining these customers therefore requires modern service processes [15].

Proactive aftersales improves customer satisfaction and retention by addressing issues before they are perceived by the customer, enabling more effective service interactions [5]. This requires tailoring services to customer needs through the integration of operational and informational data [15]. In the present context, DT-based PdM approaches support predictive interventions and improved asset lifecycle optimization, resulting in reduced costs, increased customer satisfaction, and extended asset lifetimes [21].

### E. Synthesis and Research Gap

The preceding subsections establish three building blocks: the DT as a synchronized vehicle model, PdM as a prognostic analytics layer built on top of it, and aftersales as the customer-facing business context in which service decisions are executed. Together, they form a potential end-to-end system in which the DT continuously provides component-state data to ML-based PdM models, whose forecasts are surfaced through aftersales processes to enable proactive service appointments and retain customers at the dealership.

While the referenced contributions establish the technical foundations of DT-based PdM, existing work concentrates either on algorithmic aspects in isolation [9], [13] or on industrial applications [8]. The integration of DT-based PdM into automotive aftersales business models, particularly regarding its implications for customer interaction, remains underexplored. This paper aims to address the research gap.

## III. METHODOLOGY

Building on the theoretical background established in Section II, this paper applies a *model-based gap analysis* to answer the research question. A model-based rather than experimental approach is adopted for two reasons: deploying a prototype dealership system lies outside the scope of this paper, and no public in-service wear-part failure dataset for the German passenger-car market exists at the required resolution [9]. Explicit assumptions anchored to published German market statistics therefore substitute for empirical calibration, with the trade-off that the result is a transparent, contestable estimate of potential rather than a measured outcome.

The analysis proceeds in two stages. First, a lifecycle comparison (Section IV-A) contrasts the interval-based current state against a DT/PdM-enabled target state along the customer journey, identifying the structural points at which reactive servicing causes value leakage. Second, a quantitative potential assessment translates the identified gap into economic key performance indicators (KPIs). A single economic lever—the repair capture rate $c$, defined as the share of addressable wear-part work retained at the franchised dealership—is used to quantify the gap; its target value is derived analytically from the mechanism assumptions rather than assumed directly, ensuring internal consistency. Finally, a one-way sensitivity analysis over $c$ confirms that the economic case holds across a plausible parameter band [22].

## IV. GAP ANALYSIS

### A. Current- and Target-State Lifecycle Comparison

Figure 1 illustrates the interval-based current state along the customer journey, identifying the responsible actor at each stage and where reactive servicing causes value leakage; Figure 2 presents the corresponding DT/PdM-enabled target state. Both separate the processes into a physical and a digital lane, making the extent of digital involvement directly comparable.

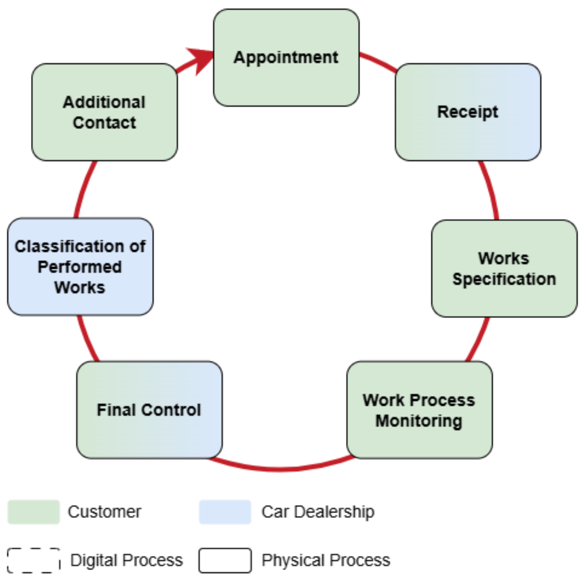


Fig. 1. Aftersales lifecycle: current interval-based state.

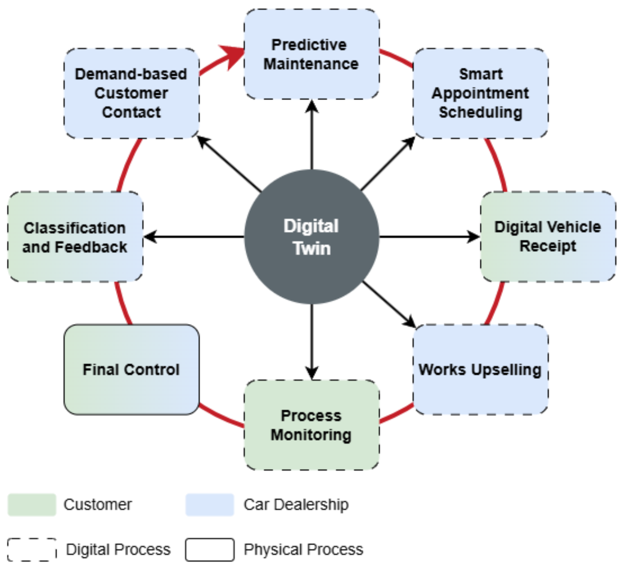


Fig. 2. Aftersales lifecycle: DT/PdM-enabled target state.

*a) Current state:* In the interval-based current state, the customer journey is driven by fixed manufacturer schedules and visible failure signals. The vehicle operates without continuous component-state monitoring; no wear forecast is computed during normal driving. Consequently, most process steps are physical in nature and therefore cannot be represented in the digital lane. The lifecycle is initiated by the customer rather than the dealership: beginning with the appointment request, the subsequent process remains largely customer-driven. Service is triggered either by a time- or mileage-based reminder or by an acute fault, such as a warning lamp. Both triggers are decoupled from the actual component condition: maintenance intervals are set conservatively, leading to premature replacement of components with substantial RUL, whereas fault-based triggers often occur only after wear limits have already been exceeded. [9].

Most process steps are controlled by the customer and only sparsely supported by digital information because no data on the actual condition of vehicle components are available. As no proactive outreach precedes a fault, customers retain complete freedom to choose a repair provider at the moment of highest urgency, causing many unplanned repairs to be captured by independent third-party workshops [17]. Such loss of control weakens the dealership's long-term customer relationship.

This leakage intensifies with vehicle age—franchised-dealer share drops from 86 % for vehicles under three years old to just 39 % for those over six years [17]—and is further reinforced by cost-driven customer behavior, with 43 % of German vehicle owners performing only the minimum required maintenance [17]. The dealership's contact with the customer is therefore reduced to the statutory inspection cycle, limiting both retention and upselling opportunities.

*b) Target state:* The DT forms the core enabler of the digital process layer. It limits direct customer involvement to the necessary interactions, thereby improving the service experience compared to the current state. The lifecycle is initiated and steered by PdM forecasts rather than customer-triggered actions. Service requests, documentation, and process steps become digitally traceable, enabling additional data-driven analysis for upselling and service revenue optimization. Nevertheless, certain physical activities, such as vehicle inspection, remain outside the scope of full digitalization.

In the DT/PdM-enabled target state, the vehicle's DT continuously aggregates sensor streams, OBD signals, operating parameters, and environmental conditions via OTA telemetry to maintain an up-to-date virtual representation of each monitored component [1]. ML models operating on this representation compute RUL for wear parts such as brake pads, discs, tires, and ignition components [13], [14]. When the forecast horizon falls below a predefined threshold, the system generates a proactive service recommendation communicated via the OEM's connected-car interface or the dealership's customer relationship management (CRM) system before any visible fault occurs [5], [21].

The appointment is scheduled at the customer's convenience, required parts are pre-ordered, and the workshop visit is transformed from an unplanned, fault-driven event into a planned, data-driven service interaction. After service completion, the DT resets the component baseline and resumes monitoring, thereby closing the lifecycle loop. The dealership thus captures wear-part replacements that would otherwise be lost to the independent aftermarket and gains valuable, structured touchpoints along the customer journey that are largely absent in the current state.

### B. Assumption Framework

Because public, in-service failure datasets representative of this setting are largely absent [9], the comparison is built on an explicit assumption framework rather than on a proprietary dataset. Assumptions are divided into *context* assumptions, which fix the reference case (Table I), *wear-part* assumptions, which size the addressable work (Table II), and *mechanism* assumptions, which govern the transition from the current to the target state (Table III). Each is anchored to a published figure where one exists and flagged as a set value otherwise.

TABLE I
CONTEXT ASSUMPTIONS.

| ID | Assumption | Value |
|---|---|---|
| A1 | Reference vehicle | Mid-segment internal combustion engine (ICE) passenger car, German market |
| A2 | Annual mileage | 13,140 km/yr [17] |
| A3 | First-owner horizon | 8 years [23] |
| A4 | Workshop visits | 1.6/yr [17] |
| A5 | Considered wear parts | Brake pads/discs, tires, wipers, 12 V battery, spark plugs |
| A6 | Service gross margin | 45 % (published range 40–55 %; set below the midpoint as a conservative estimate for mid-segment, non-premium dealers [23], [24]) |
| A7 | Franchised share | 47 % [17] |

The wear parts in A5 are deliberately limited to components whose wear is gradual, usage-driven, and not readily perceived by the driver—the class for which a forecast adds the most decision value. This class also dominates both between-interval failures and the most common defect categories in the mandatory German vehicle safety inspection (Hauptuntersuchung, HU: brakes, lighting, tires) [25]. Scheduled maintenance services (e.g., inspections and oil changes) and powertrain overhauls are out of scope. Typical replacement intervals and representative German market prices (parts plus labor, incl. VAT) are given in Table II for the reference vehicle (A1). All values are midpoints of published German market ranges [26]; actual intervals and prices vary substantially by vehicle class—compact and entry-level vehicles tend toward the lower end of each range, while premium and SUV segments tend toward the upper end. The sensitivity analysis brackets the residual uncertainty from this simplification.

TABLE II
WEAR-PART ASSUMPTIONS FOR REFERENCE VEHICLE A1 (MID-SEGMENT ICE). ALL VALUES ARE MIDPOINTS OF PUBLISHED GERMAN MARKET RANGES [26]; ACTUAL INTERVALS AND PRICES VARY BY VEHICLE CLASS.

| Part (A5) | Interval | Events/yr (at A2) | Price |
|---|---|---|---|
| Brake pads (per axle) | 40,000 km | 0.33 | € 250 |
| Brake discs (+pads, front) | 120,000 km | 0.11 | € 450 |
| Tires (set of 4) | 45,000 km | 0.29 | € 600 |
| Wiper blades (pair) | 1 year | 1.00 | € 25 |
| 12 V battery (fitted) | 6 years | 0.17 | € 150 |
| Spark plugs (set) | 60,000 km | 0.22 | € 150 |

For the mechanism, a single lever is modeled: the *repair capture rate* $c$, the share of addressable wear-part work performed at the franchised dealership rather than lost to an independent workshop. Its complement $(1 - c)$ is the leakage. The current capture rate, forecast accuracy, and proactive-offer conversion are stated in Table III; the target capture rate is derived analytically instead of being assumed directly.

Of the leaked share $(1 - c_{\mathrm{cur}})$, the fraction recaptured is the product of the forecast rate $(a)$ and the proactive-offer conversion rate $(k)$. The value $a{=}1$ is set because customers cannot distinguish a true-positive forecast from a false-positive recommendation. Every forecast therefore results in an offer that is accepted at rate $a \cdot k$, regardless of whether the underlying wear has fully materialized:

$$\begin{aligned} c_{\mathrm{tgt}} &= c_{\mathrm{cur}} + (1 - c_{\mathrm{cur}}) \cdot a \cdot k \\ &= 0.40 + 0.60 \cdot 1.0 \cdot 0.50 \\ &= 0.40 + 0.30 = 0.70. \end{aligned} \tag{1}$$

TABLE III
MECHANISM ASSUMPTIONS (CURRENT STATE → TARGET STATE).

| ID | Assumption (Symbol) | Value | Basis |
|---|---|---|---|
| A8 | Current capture rate ($c_{\mathrm{cur}}$) | 0.40 | <47 % share [17] |
| A9 | PdM forecast accuracy ($a$) | 1.0 (model) | reported 89–92 % [9]; set to 1 (see text) |
| A10 | Proactive-offer conversion ($k$) | 0.50 | Conserv. est. [5], [27] |
| — | Target capture rate ($c_{\mathrm{tgt}}$, deriv.) | 0.70 | $c_{\mathrm{cur}} + (1-c_{\mathrm{cur}}) \cdot a \cdot k$, $a{=}1$ |

This construction keeps the estimate conservative and never approaches full capture. The conversion rate ($k = 0.50$, A10) reflects this: proactive, personalized maintenance contact from the customer's own dealership is consistent with findings that proactive contact aids uptake and loyalty [5], [27]. Yet price competition from independent workshops and customer inertia toward switching service providers prevent full conversion, so the value of $k$ is treated as a deliberate floor rather than a central estimate. The value of $k$ is not empirically measured in this context; calibrating it against real customer-behavior data is identified as a direct future-work priority (Section VI-B).

### C. Potential Assessment

The calculation proceeds in three steps: the addressable wear-part value per vehicle, the captured value under each state, and the resulting KPIs at vehicle and fleet level.

*Step 1 — Addressable wear-part value:* For each part $i$, the expected annual revenue is its replacement rate $\lambda_i$ [$\mathrm{yr}^{-1}$] multiplied by its unit price $p_i$; both are from Table II, with $\lambda_i$ computed on unrounded mileage (A2).

Summing across A5 yields the annual addressable wear-part revenue per vehicle:

$$V_{\text{wear}} = \sum_i \lambda_i \cdot p_i$$
$$= 82.1 + 49.3 + 175.2 + 25.0 + 25.0 + 32.9$$
$$\approx €\,389 \quad \text{per vehicle per year.}$$

*Step 2 — Captured value:* Applying the capture rates from Table III:

$$\text{Captured}_{\text{cur}} = c_{\text{cur}} \cdot V_{\text{wear}} = 0.40 \cdot 389 \approx €\,156 \text{ /veh/yr},$$
$$\text{Captured}_{\text{tgt}} = c_{\text{tgt}} \cdot V_{\text{wear}} = 0.70 \cdot 389 \approx €\,273 \text{ /veh/yr}.$$

Applying the service gross margin (A6) gives the captured gross profit.

*Step 3 — KPIs:* Table IV summarizes the comparison. Revenue and gross-profit figures are per vehicle per year; the fleet column scales the gross-profit delta linearly to a franchised dealer group with 5,000 active customer vehicles. This size is an illustrative reference for a mid-size, multi-location dealer group rather than an empirical average; because all fleet-level figures are linear in fleet size, the results transfer proportionally to smaller or larger portfolios.

TABLE IV
CURRENT-STATE / TARGET-STATE KPI COMPARISON. DELTA FIGURES COMPUTED DIRECTLY ON UNROUNDED INTERMEDIATE VALUES; CURRENT AND TARGET CELLS ROUNDED INDEPENDENTLY.

| **KPI** | **Current** | **Target** | $\Delta$ |
|---|---|---|---|
| Repair capture rate | 40 % | 70 % | +30 pp |
| Wear-part revenue (/veh/yr) | € 156 | € 273 | +€ 117 |
| Gross profit (/veh/yr) | € 70 | € 123 | +€ 53 |
| Fleet gross profit, 5,000 veh (/yr) | € 351k | € 613k | +€ 263k |
| Customer touchpoints (/yr) | 1.6 | up to $\sim$2.7 | +$\leq$1.1 |
| First-pass inspection rate | baseline | improved | qual. |

The proactive touchpoint count rises because each high-value forecast event (the wear parts in Table II excluding the wiper case, $\approx$ 1.1 events/yr) triggers an outreach contact that did not exist in the interval-based current state. This increases the meaningful contact frequency from 1.6 to at most about 2.7 per year. The figure is an upper bound: forecast events that coincide with a scheduled service visit do not create an additional interaction. Even under substantial overlap, however, the character of the contact changes—outreach is initiated by the dealership with a concrete, vehicle-specific reason rather than a generic interval reminder—so each forecast event remains an advisory and upselling opportunity regardless of whether it adds a separate visit. The targeted parts—brakes and tires above all—are among the most common defect categories in the HU [25], so proactive replacement directly improves the customer's first-pass inspection rate. This is a tangible and communicable benefit, considering that 60 % of franchised-dealer customers currently report not being offered any additional service at their last visit [17].

*Sensitivity:* The result depends most on the target capture rate. Varying $c_{\text{tgt}}$ across a plausible band while holding $V_{\text{wear}}$ and the 45 % margin (A6) fixed gives a per-vehicle gross-profit delta of:

$$c_{\text{tgt}} = 0.50 \;\rightarrow\; \Delta \text{ gross profit} \approx €\,18 \text{ /veh/yr},$$
$$c_{\text{tgt}} = 0.60 \;\rightarrow\; \Delta \text{ gross profit} \approx €\,35 \text{ /veh/yr},$$
$$c_{\text{tgt}} = 0.70 \;\rightarrow\; \Delta \text{ gross profit} \approx €\,53 \text{ /veh/yr},$$
$$c_{\text{tgt}} = 0.80 \;\rightarrow\; \Delta \text{ gross profit} \approx €\,70 \text{ /veh/yr}.$$

Even at the conservative end, the dealership captures additional gross profit of tens of euros per vehicle per year, scaling to a six-figure annual sum across a mid-size fleet. Moreover, compounding effects are likely given that 90 % of German vehicle owners consistently return to the same workshop [17]. The economic case is therefore robust to the assumptions rather than an artifact of any single optimistic value.

### D. Limitations

The analysis is intentionally assumption-driven and carries corresponding limitations.

First, the figures are illustrative: $V_{\text{wear}}$, the capture rates, and the conversion factor are anchored to published market and accuracy data but are not fitted to an operational dataset. The contribution is therefore a transparent model of the potential, not a measured outcome. Similarly, the first-owner horizon (A3: 8 years) is an upper-bound lifecycle horizon; the DAT Report 2026 records a median holding period of 6.2 years [17], so fleet-lifetime totals are correspondingly optimistic.

Second, the reported PdM forecast accuracy ($\approx$90 %) is stated for transparency but does not enter the capture-rate formula. A proactive service recommendation looks identical to the customer regardless of whether the underlying forecast is correct or slightly premature. Both true-positive and false-positive predictions therefore generate the same type of offer, which is accepted at rate $k$. Accuracy will improve as in-service data accumulates [9], but the model outcome does not depend on reaching any particular accuracy threshold.

Third, and most consequential for practice, the target state presumes the dealership can actually obtain the vehicle's telemetry. The vehicle owner must authorize a third party such as the dealership to receive in-vehicle data, while the OEM remains responsible for providing that data in a usable format. Access is thus contingent on customer consent, OEM provisioning, and business-to-business compensation—a non-technical adoption barrier that the model does not price in.

## V. Discussion and Recommendations

### A. Interpretation of Results

The per-vehicle gross-profit uplift of € 53 per year scales to € 263k annually across a 5,000-vehicle fleet and compounds to approximately € 2.1 million over the optimistic eight-year first-owner horizon (A3). The parts margin is, however, only the visible component of the value. Each of the up to 1.1 additional contact events per vehicle per year is simultaneously an advisory opportunity and a loyalty-reinforcing interaction, consistently linked to increased repurchase intent at the originating dealership [5], [27]. The sensitivity analysis confirms robustness: even at $c_{\mathrm{tgt}} = 0.50$, the fleet-level gain remains approximately € 90k per year—a positive business case under any plausible conversion scenario.

The model deliberately quantifies only the benefit side: investment costs for the DT platform, CRM integration, and consent infrastructure are OEM- and dealer-specific and not publicly available. The gross-profit delta nevertheless defines a break-even threshold. At € 53 per vehicle per year, the system pays off as long as its total running cost remains below roughly € 4.40 per connected vehicle per month. This gives practitioners a concrete benchmark against vendor pricing rather than a speculative return-on-investment figure.

### B. The Non-Technical Barrier

The DT/ML/OTA technology stack is commercially mature and not the binding constraint [1], [9]. The real bottleneck is the telemetry access: as outlined in Section IV-D, in-vehicle data reaches the dealer only with the owner's consent and on OEM-provisioned terms, making the dealer dependent on two actors it does not control. The practical recommendation is therefore to build consent infrastructure—opt-in flows integrated into the CRM and clear customer value propositions—as a prerequisite rather than an afterthought. This holds independently of any specific DT deployment decision.

### C. Implications for Practice and Research

Practitioners should position DT-based PdM as a CRM extension rather than an IT project. The measurable value driver is the quality of proactive outreach, not forecast accuracy [5], [21]; appointment-scheduling integration and communication design should therefore precede algorithm tuning. A phased entry via OEM brands that already expose connected-car telemetry APIs allows conversion rates to be measured before a broader infrastructure commitment is made.

For researchers, the priority is the empirical validation of the conversion rate $k$, which is modeled here but not fitted to operational data. A field study measuring opt-in and outreach acceptance rates would directly calibrate this parameter.

Beyond the dealership's business case, DT-based PdM exemplifies the shift from merely smart to wise information systems. Replacing parts on evidence instead of fixed intervals avoids premature replacement and the associated material waste. The explicit assumption framework keeps the decision model transparent and contestable, while consent-based data access keeps the customer in control of the decision loop.

## VI. Conclusion and Future Work

### A. Conclusion

This paper examined the effect of digital-twin-based predictive maintenance on automotive aftersales, focusing on the German passenger-car market. A lifecycle comparison showed that the transition from interval-based servicing to a DT/PdM-enabled target state redirects wear-part work from the independent aftermarket back to the franchised dealership by replacing reactive, fault-driven appointments with proactive, evidence-based service recommendations. A quantitative potential assessment translated this shift into concrete KPIs: under a conservative conversion assumption ($k = 0.50$), the repair capture rate rises from 40 % to 70 %, increasing gross profit by approximately € 53 per vehicle per year and € 263k annually across a 5,000-vehicle fleet. Customer touchpoints increase by up to 1.1 per year, from 1.6 to roughly 2.7, each an additional advisory and upselling opportunity. The dominant adoption barrier is non-technical: access to in-vehicle telemetry requires customer consent and OEM provisioning, a hurdle any real-world deployment must resolve.

In answer to the research question: the effect of DT-based PdM on automotive aftersales is a measurable shift of wear-part revenue from the independent aftermarket to the franchised dealership, accompanied by a structural increase in customer contact frequency that amplifies retention and upselling value beyond the parts margin alone.

### B. Future Work

Three directions stand out for future research. First, the assumption framework should be validated against an operational DT deployment, replacing modeled KPIs with measured outcomes and calibrating the conversion rate $k$ under real customer-behavior conditions. Second, the data-access challenge warrants dedicated study: viable consent architectures and OEM-dealer data-sharing agreements need to be developed, and their effect on customer opt-in quantified. Third, the framework should be extended to battery electric vehicles, where the high-voltage battery's state of health replaces spark plugs as the dominant prognostic target.

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## Verwendung Generativer KI-Systeme

Bei der Erstellung der eingereichten Arbeit haben wir die nachfolgend aufgeführten auf künstlicher Intelligenz (KI) basierten Systeme benutzt:

1) Claude (Anthropic)
2) ChatGPT (OpenAI)

Wir erklären, dass wir

- uns aktiv über die Leistungsfähigkeit und Beschränkungen der oben genannten KI-Systeme informiert haben,
- die aus den oben angegebenen KI-Systemen direkt oder sinngemäß übernommenen Passagen gekennzeichnet haben,
- überprüft haben, dass die mithilfe der oben genannten KI-Systeme generierten und von uns übernommenen Inhalte faktisch richtig sind,
- uns bewusst sind, dass wir als Autorinnen bzw. Autoren dieser Arbeit die Verantwortung für die in ihr gemachten Angaben und Aussagen tragen.

Die oben genannten KI-Systeme haben wir wie im Folgenden dargestellt eingesetzt:

**Tabelle I. Einsatz von KI-Systemen**

| **Arbeitsschritt** | **Eingesetzte(s) KI-System(e)** | **Beschreibung der Verwendungsweise** |
|---|---|---|
| Formatierung und Satz | Claude (Anthropic) | Umwandlung von Fließtext in LaTeX (Tabellen, Formeln, Literaturverzeichnis) |
| Verifikation der Berechnungen | Claude (Anthropic) | Prüfung der Zwischenrechnungen ($V_{\mathrm{wear}}$, Capture-Raten) auf Konsistenz |
| Sprachliche Überarbeitung | Claude (Anthropic), ChatGPT (OpenAI) | Korrektur von Grammatik und Stil einzelner Abschnitte |